ALEC HUDSON

ahudson05@me.com LinkedIn - AHudson AHudson Projects Portfolio 615-638-7771

EDUCATION

Auburn University, Samuel Ginn College of Engineering, Auburn, AL
Bachelor of Science Mechanical Engineering (BSME), **Expected Graduation: Fall 2027**

- GPA: 3.99 | Honors College | Dean's List
- Member: Tau Beta Pi National Engineering Honor Society
- Member: Pi Tau Sigma Mechanical Engineering National Honor Society
- Relevant Coursework – Kinematics and Dynamics, Thermodynamics I & II, System Dynamics and Controls, System Modeling & Vibration Analysis, Mechanics of Materials, Basic Composites, Mechanical Engineering Design, Computer Aided Engineering, Fluid Mechanics

TECHNICAL: SKILLS	**3D Printing**, **CNC**, **Laser Cutter** expertise, **Arduino Microcontroller and IDE** electronic programming and prototyping, Finite Element Analysis, Mechanical aptitude: design, manufacture & assembly, Strong analysis & calculation skills, Vibration Analysis and Stability Optimization, Financial Analysis
SOFTWARE: PROFICIENCY	Advanced **MATLAB/Simulink** skills**,** Advanced **SolidWorks**/**Fusion modeling**/drawing, **Python**, **Excel & MS Project**, **Power Apps**, **Tableau** Data Visualization, **JIRA** & **Confluence** proficient, **Unreal Engine**, **Blender** Modeling Expertise, **G-code**, Adobe **Photoshop**, **Premiere**, **After Effects**
ADDITIONAL: SKILLS	**Gemma and Qwen LLM AI** configuration and training, **ASTM-F24 2291** Familiarity, Project Management/Entrepreneur in high pressure environments, Hands-on experience building, prototyping & testing mechanisms, machines & assemblies

ENGINEERING WORK EXPERIENCE

Walt Disney Imagineering Intern – Ride Development **June 2026 – December 2026**

- Contributed to the design, installation and testing of a new theme park attraction with a multidisciplinary ride team; authored test plans and BOMs and collaborated with show, controls and vendor partners.
- Designed a sliding access door and gate assembly using aluminum extrusion framing and detent hold-open features; led trade studies comparing swinging and sliding concepts for clearance, safety and maintenance.
- Performed **FEA**, hand calculations and swept-motion interference studies in **SolidWorks**; designed 3D-printed and sheet metal enclosures and brackets for operator controls and sensors.
- Developed an automated shipping log and parts tracking system using **Excel**, **HTML** and **AI tools**, reconciling bills of lading against BOMs for real-time visibility of incoming ride components.

Walt Disney World Mechanical Engineering Intern – Textile Services **January 2026 – June 2026**

- Managed and delivered **30** concurrent engineering projects across all 4 Disney textile plants, applying **3D modeling**, mechanical analysis and clear communication with operations and engineering teams.
- Challenged the initial engineering proposal for a mechanical lift in a linen pit; designed and implemented a simplified static platform that reduced complexity, eliminated failure modes and decreased stress and strain on laundry bags and Cast Members.
- Designed and **3D modeled** facility components for easier manufacturing, including a fabricated rail guide that prevented a month of downtime and is used **1,900** times a day.

Auburn University, Tutor – System Dynamics and Controls **August 2025 – December 2025**

- Led group tutoring sessions for **30+** students in System Dynamics and Controls, covering dynamic system modeling, controllers and vibration.
- Partnered with professors to align sessions with weekly course topics, using practical examples in statics, dynamics and kinematics.

Auburn University, Design & Innovation Center **September 2023 – Present**

- Train and supervise students on engineering equipment as a volunteer Makerspace Assistant (4–6 hours per week).
- Support mechanical and electrical prototyping, **3D printing** and **G-code** for **CNC** manufacturing, helping students fabricate high-quality project components.

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WORK EXPERIENCE

The Seal Team |Service Guys, Brentwood, TN, Founder **April 2015 – Present**

- Founded and led a power washing, driveway repair and sealing business; assemble, maintain and troubleshoot equipment, building hands-on skills with mechanisms and machines.
- Developed custom pricing, product volume and same-day estimating methods to improve operational efficiency.
- Manage payroll, scheduling and a team for lawn care and maintenance of **24** residential properties.

PROJECTS

Edge-AI Video Analytics System

- **Designed and deployed a fully local, multi-camera AI video analytics system** on an **NVIDIA DGX Spark**, built to scale to 16+ IP camera streams with no cloud dependency.
- Integrated **YOLO** detection with **TensorRT**, multi-object tracking with re-identification, license plate recognition and the **Qwen3-VL** vision-language model to describe and search events in natural language.
- Architected a **Dockerized microservice** stack with **PostgreSQL/pgvector** semantic search, continuous recording and a real-time **WebRTC** dashboard with PTZ camera controls.

Autonomous Stair-Climbing Robot

- **Led a multidisciplinary team** to design and prototype a self-navigating robot for stairs and flat surfaces, integrating mechanical assemblies, motors and power management.
- Created optimized 3D CAD models in **SolidWorks** for **CNC** and **water-jet** fabrication, ensuring efficient assembly and structural integrity.
- Directed manufacturing, assembly and iterative testing through three design revisions, exceeding required performance by **900%**.

Autonomous Steering Controller for IndyCar Simulation

- Modeled vehicle steering dynamics with a bicycle model in **MATLAB** to simulate realistic behavior and predict lateral forces.
- Designed and implemented **P, PI and PID controllers** using waypoints for precise trajectory tracking and reduced tracking error.
- Applied vibration analysis and tuned controller parameters to minimize lateral deviation, improving vehicle stability and simulation accuracy by a factor of 10.

IV Bag with Position Control and Audible Alerts

- Designed and prototyped an automated IV bag lifting mechanism using an electric linear actuator, with real-time height feedback, fluid level monitoring to **0.1%** accuracy and user alerts, achieving **100%** reliability.
- Engineered the control system with an **Arduino Microcontroller** and **Motor Shield** in **Arduino IDE**, managing position control, audible alerts, motor actuation and load cell-based mass measurement.
- Established design specifications and project scope, then sourced and assembled off-the-shelf and **3D-printed** components into a fully functional prototype.

AWARDS AND ASSOCIATIONS

- Earned 2nd Place in Intro to Mechanical Engineering Design for Pneumatic Powered Vehicle (2023)
- Awarded Rookie of the Year for Auburn Marching Band Percussion (2023)
- Awarded Citizenship Leader Award JROTC Battalion, Ravenwood High School (2023)